

Embracing 5G, Shaping the Digital Future



IR925 Industrial Router

· 5G

· Wi-Fi 6

· Security

· Cloud
Management

1. Product Overview

The InRouter925 (IR925) series is an industrial-grade router designed by InHand Networks to enable enterprise digital transformation. Integrating 5G cellular connectivity, Wi-Fi, and VPN technologies, the IR925 delivers efficient, secure, and intelligent networking solutions for a wide range of IoT applications. Its multi-level link detection mechanism ensures stable and reliable communications, fully meeting the requirements of unattended industrial sites. When paired with the InHand DeviceLive cloud management platform, the IR925 enables centralized cloud-based management and remote maintenance, allowing users to monitor device status effortlessly, reduce deployment costs, and improve operational efficiency.

1.1 Key Features and Benefits

Value Proposition	Description
5G	5G cellular connectivity with downlink speeds up to 3.4 Gbps; backward-compatible with 4G/3G; high bandwidth and low latency for next-generation industrial networking
Wi-Fi 6	Dual-band concurrent throughput up to 3000 Mbps (2.4 GHz & 5.8 GHz); 802.11 ax/ac/a/b/g/n; AP/Client dual-mode; Wi-Fi captive portal and multi-SSID support
Always-On Connectivity	Link backup, link detection, dual-SIM failover, and embedded hardware watchdog ensure reliable and stable connections with minimized risk of network outages
Security	Multi-layered security including VPN, firewall, and policy-based routing; supports IPsec/L2TP/OpenVPN/WireGuard; TPM hardware security module
Cloud Management	Integration with DeviceLive platform for centralized management of tens of thousands of distributed sites; remote connectivity, diagnostics, debugging, and remediation reduce on-site maintenance costs

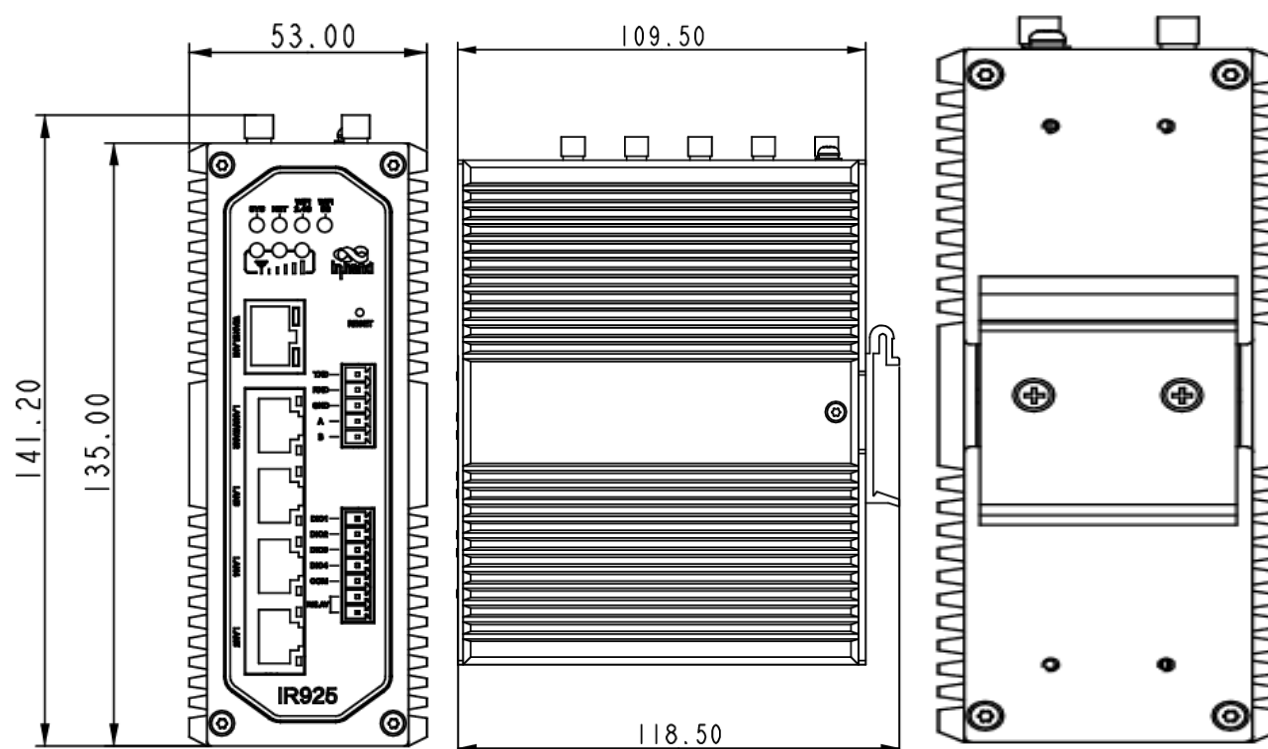
Value Proposition	Description
Industrial-Grade Design	Metal enclosure, fanless cooling, IP30-rated; operating temperature -40°C to $+70^{\circ}\text{C}$; DC 9–48 V wide-range input; DIN-rail or wall-mount installation

1.2 Application Scenarios

Scenario	Typical Customers	Key Requirements	Value Proposition
Smart Factory	Manufacturing enterprises, system integrators	Production-line equipment connectivity, remote O&M, high-availability links	5G/wired multi-link access to support 24/7 continuous production
Warehouse Automation	Logistics & warehousing, automation integrators	AGV/AMR connectivity, on-site wireless coverage, centralized management	Wi-Fi 6 with cellular backup to improve operational efficiency and reduce labor costs
Telemedicine	Hospitals, medical device manufacturers	Remote consultation, secure medical data transmission	High-bandwidth, low-latency connectivity to extend the reach of specialist care
Inspection Robots	Power, petrochemical, manufacturing inspection	Real-time video and telemetry backhaul, mobile always-on connectivity	Dual SIM and link detection for rigorous monitoring of production processes

Scenario	Typical Customers	Key Requirements	Value Proposition
Distribution Automation	Utilities, energy service providers	Rapid alarm transmission, secure isolated communications	VPN and multi-layer firewall for faster, more accurate grid control
Smart City	Municipal authorities, campus/city operators	Distributed site connectivity, unified cloud management	DeviceLive centralized management for secure, sustainable urban connectivity

1.3 Product Dimensions



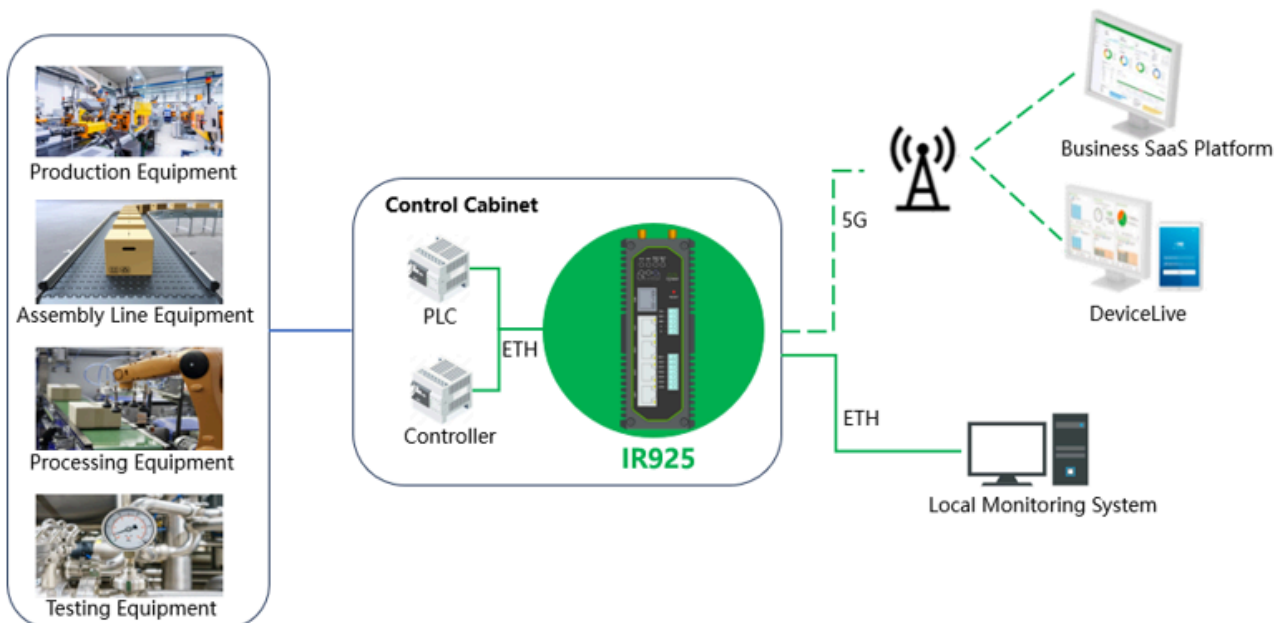
Notes:

1. All dimensions are in millimeters.
2. Dimensions (L × W × H): 141.2 × 118.5 × 53 mm.
3. All dimensions are approximate and for reference only.

2. System Architecture

2.1 Solution Components

The IR925 solution comprises the industrial router unit and external antennas, and interfaces with local monitoring systems and the DeviceLive cloud platform for on-site device connectivity and remote O&M.



2.2 Deployment

Typical single-site standard configuration (subject to adjustment based on model and site requirements):

Component	Quantity	Description
IR925 Unit	1	DIN-rail or wall-mount, deployed in cabinet or field enclosure

Component	Quantity	Description
5G Antennas	4	4 × 5G antennas
Wi-Fi Antennas	2	2 × Wi-Fi antennas
GNSS Antenna	1	1 × GNSS antenna
SIM Cards	2	Drawer-type slot, dual Nano-SIM support
Power Supply	1	DC 9–48 V, 2-pin industrial terminal block, overcurrent and reverse-polarity protection
Ethernet Ports	5	RJ45: 1 × 2.5GbE + 4 × 1GbE, WAN/LAN/VLAN configurable

3. Hardware Specifications

3.1 Hardware Overview

Parameter	Specification
Processor	1 GHz
Memory	512 MB
Storage	8 GB eMMC
Cellular	5G NR, downlink up to 3.4 Gbps, Sub-6 GHz (450 MHz – 6 GHz); backward-compatible with 4G/3G (see ordering information for supported bands)
SIM Slot	1 × drawer-type slot, dual Nano-SIM, 1.8 V / 3 V; eSIM (optional)
Ethernet Ports	1 × 2.5GbE + 4 × 1GbE RJ45, WAN/LAN/VLAN configurable; 1.5 kV network isolation transformer protection

Parameter	Specification
Serial Ports	1 × RS232, 1 × RS485; ESD protection: 15 kV
I/O	4 × DIO (DO/DI configurable); 1 × relay, 2 A @ 30 V DC
Wi-Fi	2.4 GHz & 5.8 GHz, dual-band concurrent up to 3000 Mbps; 802.11 ax/ac/a/b/g/n (WLAN model)
Wi-Fi Tx Power	5 GHz: 21 dBm; 2.4 GHz: 21 dBm
Wi-Fi Range	Up to 50 m line-of-sight (actual range depends on environment)
Antenna Connectors	4 × 5G or 2 × 4G; 2 × Wi-Fi; 1 × GNSS
LED Indicators	1 × System, 1 × Cellular, 1 × 2.4G Wi-Fi, 1 × 5G Wi-Fi, 3 × Signal
Grounding Terminal	Supported
Hardware Components	RTC supported; TPM supported
Power Input	DC 9–48 V, overcurrent protection, reverse-polarity protection, 2-pin industrial terminal block
Reset Button	Pinhole reset
Enclosure	Metal
Cooling	Fanless
Ingress Protection	IP30
Operating Temperature	–40° C to +70° C

Parameter	Specification
Storage Temperature	-40° C to +85° C
Operating Humidity	5% - 95% (non-condensing)
Dimensions	141.2 × 118.5 × 53 mm
Weight	1002 g
Mounting	DIN-rail, wall-mount

3.2 Interface Overview

Ethernet Ports (× 5)

- 1 × 2.5GbE + 4 × 1GbE RJ45
- WAN/LAN/VLAN configurable
- 1.5 kV network isolation transformer protection

Serial Ports

- 1 × RS232, 1 × RS485
- ESD protection: 15 kV

I/O and Relay

- 4 × DIO (DO/DI configurable)
- 1 × relay, 2 A @ 30 V DC

Power / Antenna / SIM

- DC 9-48 V, 2-pin industrial terminal block, overcurrent and reverse-polarity protection
- Antennas: 4 × 5G; 2 × Wi-Fi; 1 × GNSS
- Drawer-type dual Nano-SIM; eSIM optional
- Pinhole reset button; grounding terminal supported

3.3 Environmental and Reliability Standards

Test Item	Standard	Level / Result
ESD (Electrostatic Discharge)	EN 61000-4-2	Level 3
Radiated Immunity	EN 61000-4-3	Level 3
EFT (Electrical Fast Transient)	EN 61000-4-4	Level 3
Surge	EN 61000-4-5	Level 3
Conducted Immunity	EN 61000-4-6	Level 3
Oscillatory Wave	EN 61000-4-12	Level 3
Power Frequency Magnetic Field	EN 61000-4-8	Horizontal/Vertical 400 A/m ($\geq$ Level 2)
Shock	IEC 60068-2-27	Compliant
Vibration	IEC 60068-2-6	Compliant
Free Fall	IEC 60068-2-32	Compliant

3.4 Key Hardware Advantages

Advantage	Description
Multi-Rate Ethernet	1 × 2.5GbE + 4 × 1GbE to accommodate high-bandwidth field devices and multi-terminal access
Comprehensive Industrial Interfaces	RS232/RS485, 4 × DIO, and relay output for seamless integration with industrial control systems and field devices

Advantage	Description
Wide Temperature & Voltage Range	–40° C to +70° C operating temperature, DC 9–48 V power input for harsh industrial environments
Multi–SIM Support	Dual Nano–SIM link switching with eSIM footprint reserved to enhance cellular connectivity availability
TPM + Fanless Design	Hardware TPM support; fanless passive cooling for long–term unattended operation
Flexible Mounting	DIN–rail and wall–mount options for easy deployment in cabinets and enclosures

4. Software Features

4.1 Network Protocols

Category	Supported Protocols
Network Access	APN, VPDN
Authentication	CHAP, PAP
Network Types	WCDMA, TDD LTE / FDD LTE, 5G NR (SA/NSA) (see ordering information for supported bands)
LAN Protocols	ARP, Ethernet
WAN Protocols	Static IP, DHCP, PPPoE
Routing	Static routing, OSPF, BGP

Category	Supported Protocols
Protocols & Applications	TCP, UDP, IPv4, IPv6, ICMP, NTP, DNS, HTTP, HTTPS, SSL/TLS, ARP, VRRP*, PPPoE, SSH, DHCP Server, DHCP Relay, DHCP Client, DDNS, Telnet, IP Passthrough

4.2 Network Security

Security Feature	Description
Firewall	MAC/IP/port/protocol-based filtering; NAT and port mapping; access control; policy-based routing; 802.1X
Data Security	IPsec VPN, L2TP VPN, OpenVPN, WireGuard VPN, VXLAN*; CA certificate support
Management Security	Role-based access control with privilege levels

4.3 Link Management and Backup

Failover

- Interface backup, VRRP*

Dual-SIM Switching

- Dual-SIM link failover

Link Keepalive

- Heartpack-based detection with automatic reconnection

Embedded Watchdog

- Self-diagnostic monitoring with automatic fault recovery

4.4 Wi-Fi Features

Operating Modes



- AP and Client dual-mode

Security

- Open system, shared key, WPA/WPA2/WPA3 authentication
- WEP/TKIP/AES encryption

Additional Features

- Wi-Fi captive portal
- Multi-SSID support

4.5 GNSS

- GNSS IP forwarding and serial forwarding
- Time synchronization

4.6 DTU Function

- TCP/UDP transparent transmission
- Modbus RTU to Modbus TCP conversion

4.7 Monitoring, Alarm, and QoS

Dashboard

- Device information, interface status, granular traffic statistics

Link Monitoring

- Link latency, jitter, packet loss, and throughput monitoring

Cellular Signal

- Real-time cellular signal metrics: RSSI, RSRP, RSRQ, SINR

Logging and Alarms

- System logs, diagnostic logs, device events, email alarms

QoS

- Traffic shaping



4.8 System Management

Remote Access

- Web and CLI remote access

Network Management

- InHand DeviceLive cloud NMS platform with batch management and remote maintenance
- SNMP v1/v2c/v3

SMS

- SMS-based device reboot

Network Diagnostics

- Ping, Traceroute, packet capture, speed test

Configuration Backup

- Configuration import/export
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5. Cloud Management

5.1 DeviceLive Platform Integration

The IR925 integrates with the InHand DeviceLive device management and operations platform, enabling centralized management of tens of thousands of distributed site devices through a unified cloud interface. Users can remotely connect to field devices via cloud services for diagnostics, debugging, and remediation, reducing losses caused by device failures and making management easier, more efficient, and more cost-effective.

5.2 Cloud Management Features

Feature	Description
Real-Time Monitoring	Monitor field device status anytime, anywhere
Batch Management	Batch manage distributed site devices through DeviceLive
Remote Maintenance	Remote maintenance to reduce on-site O&M costs
Remote Connectivity	Cloud-based remote connection to field devices for diagnostics, debugging, and remediation

6. Ordering Information

6.1 Model Number Structure

Model	Region	Supported Bands
IR925-CNNR-WLAN	China	5G NR NSA: n41/78/79 5G NR SA: n1/3/5/8/28A/41/77/78/79 LTE FDD: B1/3/5/8 LTE TDD: B34/38/39/40/41 WCDMA: B1/5/8
IR925-GLNR-WLAN	North America	5G NR NSA: n1/2/3/5/7/8/12/13/14/18/20/25/26/28/29/30/38/40/41/48/66/70/71/75/76/77/78/79 5G NR SA: n1/2/3/5/7/8/12/13/14/18/20/25/26/28/29/30/38/40/41/48/66/70/71/75/76/77/78/79 LTE FDD: B1/2/3/4/5/7/8/12/13/14/17/18/19/20/25/26/28/29/30/32/66/71 LTE TDD: B34/38/39/40/41/42/43/48 LTE LAA: B46 WCDMA: B1/2/4/5/8/19
IR925-EUNR-WLAN	Europe / Asia-Pacific	5G NR NSA: n1/3/7/28/38/40/41/77/78 5G NR SA: n1/3/5/7/8/20/28/38/40/41/66/77/78 LTE FDD: B1/2/3/4/5/7/8/20/28/66 LTE TDD: B38/40/41 LTE LAA: B46 WCDMA: B1/2/5/8

Note: The model number IR925–CNNR–WLAN denotes the IR925 series wireless router with 5G/4G cellular connectivity, Wi-Fi 6 with AP & Client modes, RS232 and RS485 serial interfaces, digital I/O, and GPS.

7. Installation and Deployment

7.1 Mounting Options

DIN–Rail Mount

- DIN–rail mounting for industrial cabinet deployment

Wall Mount

- Wall–mount installation for enclosure and wall fixation

7.2 Installation Steps

Step	Task
1	Select DIN–rail or wall–mount installation based on site requirements and install the IR925 unit
2	Connect external antennas (5G/4G, Wi-Fi, GNSS as per model configuration)
3	Insert SIM cards (install both SIMs for dual–SIM configuration as needed)
4	Connect power supply (DC 9–48 V) and ensure proper grounding
5	Connect Ethernet, serial, and I/O devices as required
6	Power on and complete initial configuration via Web or CLI
7	(Optional) Connect to the DeviceLive cloud platform

7.3 Installation Precautions

- Ensure optimal antenna placement for best signal reception

- Verify power supply voltage (9–48 V DC) before connection
 - Properly ground the device
 - Route cables away from strong interference sources and heat sources
 - Verify SIM card orientation before insertion
 - After initial power-up, complete basic network security configuration (firewall / VPN, etc.)
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8. Contact Us

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